12-11 第一篇

Telescopes and Astronomy: Exploring the Universe

The telescope was invented over four hundred years ago and changed how we see the universe. Before telescopes, people could only observe the sky with their eyes. With a simple tube and lenses, astronomers discovered moons around Jupiter, spots on the Sun, and countless stars that were invisible before. The telescope opened a window to the cosmos. [1]

Telescopes work by collecting and focusing light. Optical telescopes use lenses or mirrors to gather light from distant objects and make them appear larger and brighter. The larger the lens or mirror, the more light it collects and the farther we can see. Today, some telescopes have mirrors tens of metres across. They can observe galaxies billions of light-years away. [2]

Space telescopes have pushed our view even farther. The Hubble Space Telescope, launched in 1990, orbits above Earth's atmosphere. Without air to blur the view, it has captured stunning images of nebulae, galaxies, and distant planets. The James Webb Space Telescope, launched in 2021, can see even deeper into the early universe. These instruments reveal secrets that ground-based telescopes cannot reach. [3]

Astronomy helps us understand our place in the universe. We have learned that Earth is one of billions of planets. We have found evidence of water on Mars and moons that might harbour life. Telescopes have also helped confirm ideas about the Big Bang and dark matter. Each discovery raises new questions and deepens our curiosity. [4]

Telescopes and astronomy continue to advance. New technology allows us to detect gravitational waves, photograph black holes, and study the atmospheres of distant planets. Whether you use a small telescope in your backyard or follow the latest space missions online, the universe is more accessible than ever. [5]

选项

A. In this way, the telescope became one of the most important tools in the history of science. B. This principle allows astronomers to see objects that are too far or too faint for the naked eye. C. As a result, space telescopes have transformed our understanding of the cosmos. D. Through these discoveries, we learn how small we are—and how much there is still to explore. E. For these reasons, astronomy remains an exciting and evolving field for anyone curious about the stars. F. For example, the Andromeda Galaxy can be seen with the naked eye as a faint smudge on a dark night.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前介绍望远镜发明四百余年、此前只能肉眼观天、简单镜筒与透镜使天文学家发现木卫、太阳黑子与无数此前不可见之星、为宇宙打开窗口，空后需总结望远镜在科学史上的地位。A的"In this way"承接前文，"one of the most important tools in the history of science"与前文对应。
- **[2] B**：
  - **线索**：空前说望远镜通过收集与聚焦光线工作、光学望远镜用透镜或反射镜聚光、镜面越大收集光越多看得越远、现代大型望远镜口径达数十米、可观测数十亿光年外的星系，空后需总结原理带来的观测能力。B的"This principle"指代前文，"see objects that are too far or too faint for the naked eye"与前文呼应。
- **[3] C**：
  - **线索**：空前说空间望远镜拓展视野——哈勃1990年发射、轨道不受大气干扰、拍摄星云星系与遥远行星、韦伯2021年发射、可观测更早宇宙、揭示地面望远镜无法触及的秘密，空后需总结空间望远镜对认知的变革。C的"As a result"承接前文，"space telescopes have transformed our understanding of the cosmos"与前文对应。
- **[4] D**：
  - **线索**：空前说天文助我们理解宇宙中的位置、地球只是数十亿行星之一、发现火星水与可能有生命的卫星、验证大爆炸与暗物质、每项发现引发新问题加深好奇心，空后需总结发现带来的渺小感与探索空间。D的"Through these discoveries"指代前文，"how small we are—and how much there is still to explore"与前文呼应。
- **[5] E**：
  - **线索**：空前说望远镜与天文持续进步、新技术可探测引力波拍摄黑洞研究系外行星大气、从后院小望远镜到在线追航天任务、宇宙比以往更可及，空后需总结天文对好奇者的吸引力。E的"For these reasons"承接前文，"astronomy remains an exciting and evolving field for anyone curious about the stars"与前文对应。
- **干扰项分析**：
  - **F. For example, the Andromeda Galaxy can be seen with the naked eye...**：以For example开头，为肉眼可见天体案例（仙女座星系）。各空均为概括

或过渡句；[2]易被误选因涉及肉眼与望远镜对比，但该空需总结望远镜原理带来的观测能力，而非补充肉眼可见目标的举例。